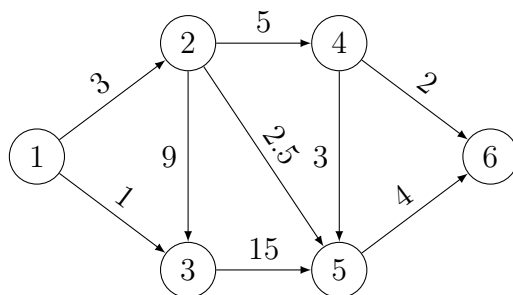


Dijkstra's algorithm

Michel Bierlaire

Solution of the practice quiz

1. Network 1:



We first note that all arcs in the network have non negative cost, so that Dijkstra's algorithm can be applied. It is the same algorithm as the shortest path algorithm, with two differences:

- the node that is treated at each iteration is the node in $\mathcal{S}$ with the smallest label,
- there is no need to identify a cycle with negative cost, as all arcs have non negative cost.

The iterations of Dijkstra's algorithm are reported in Table 1, where

- the first column reports the number of the iteration,
- the second column reports the set $\mathcal{S}$ of nodes to be treated at the beginning of the iteration,
- the third column reports the node that is treated during the iteration,

Iteration	S	i	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
0	$\{1\}$	1	0	∞	∞	∞	∞	∞
1	$\{2, 3\}$	3	0	3	1	∞	∞	∞
2	$\{2, 5\}$	2	0	3	1	∞	16	∞
3	$\{4, 5\}$	5	0	3	1	8	5.5	∞
4	$\{4, 6\}$	4	0	3	1	8	5.5	9.5
5	$\{6\}$	6	0	3	1	8	5.5	9.5
6	$\{\}$	-	0	3	1	8	5.5	9.5

Table 1: Iterations of Dijkstra's algorithm for Network 1

- the remaining columns report the values of the labels at the beginning of the iteration.

In order to check how the algorithm works, we focus on the third iteration. At the beginning of iteration 3, both nodes 4 and 5 are in set S . As $\lambda_4 = 8$ and $\lambda_5 = 5.5$, it is decided to treat node 5, as it has the smallest label.

The shortest path spanning tree is reported in Figure 1. Finally, we observe the invariants of the algorithm for iteration 4. At the beginning of the iteration, the set of permanent labels is

$$\mathcal{T} = \{1, 2, 3, 5\}$$

and the set of nodes to be treated is

$$\mathcal{S} = \{4, 6\}$$

We verify the invariants of the algorithm:

- During iteration 4, only the label of node 6 is updated. And no node in $\mathcal{T}$ is updated.
- No node in $\mathcal{T}$ is included in $\mathcal{S}$ during the iteration. Indeed, after the iteration, $\mathcal{S} = \{6\}$.
- The labels of any node i in $\mathcal{T}$ corresponds to the length of a shortest path from the origin to i . For instance, $\lambda_5 = 5.5$ corresponds to the length of path $1 \rightarrow 2 \rightarrow 5$, that is the shortest path between 1 and 5.

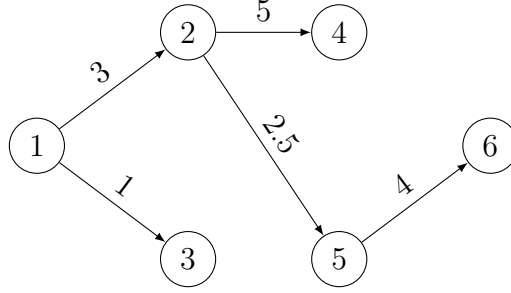
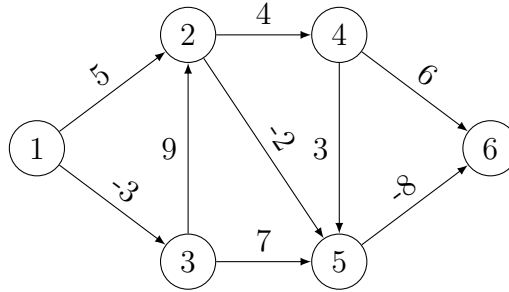


Figure 1: Shortest path spanning tree for Network 1

- The labels of nodes 4 and 6 (nodes not in $\mathcal{T}$) are larger than the labels of any node in $\mathcal{T}$.
2. Network 2: A condition to apply Dijkstra algorithm is that all costs are positive. Here it's not the case. Therefore, we apply the shortest path algorithm using the same rule as Dijkstra's algorithm for the selection of the node to process at each iteration.



The iterations of the shortest path algorithm are reported in Table 2, and the shortest path spanning tree in Figure 2. Note that the lower bound is

$$(m - 1) \min_{(i,j)} c_{ij} = 5 \cdot -8 = -40.$$

The algorithm converges without reaching the lower bound. There is no negative costs cycle. It is important to note that, despite the use of Dijkstra's rule for the selection of the node to process, the algorithm does not have the property that each node is processed at most once.

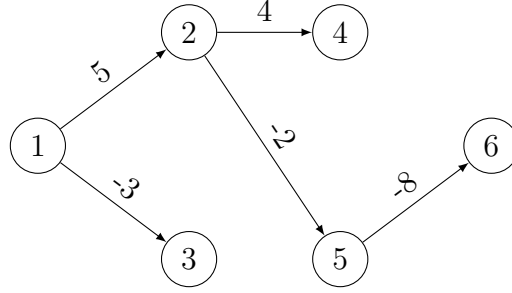


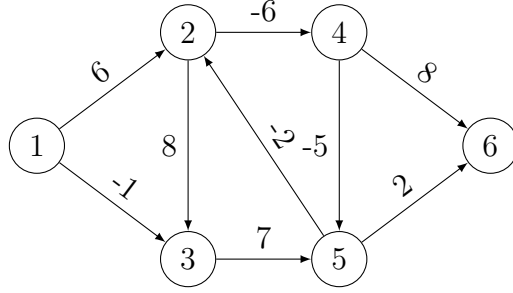
Figure 2: Shortest path spanning tree for Network 2

In particular, nodes 5 and 6 have been processed twice during the algorithm.

Iteration	S	i	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
0	{1}	1	0	∞	∞	∞	∞	∞
1	{2, 3}	3	0	5	-3	∞	∞	∞
2	{2, 5}	5	0	5	-3	∞	4	∞
3	{2, 6}	6	0	5	-3	∞	4	-4
4	{2}	2	0	5	-3	∞	4	-4
5	{4, 5}	5	0	5	-3	9	3	-4
6	{4, 6}	6	0	5	-3	9	3	-5
7	{4}	4	0	5	-3	9	3	-5
8	{}	—	0	5	-3	9	3	-5

Table 2: Iterations of the shortest path algorithm for Network 2

3. Network 3: A condition to apply Dijkstra algorithm is that all costs are positive. Here it's not the case. Therefore, we apply the shortest path algorithm using the same rule as Dijkstra's algorithm for the selection of the node to process at each iteration.



Iteration	S	i	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
0	{1}	1	0	∞	∞	∞	∞	∞
1	{2, 3}	3	0	6	-1	∞	∞	∞
2	{2, 5}	2	0	6	-1	∞	6	∞
3	{4, 5}	4	0	6	-1	0	6	∞
4	{5, 6}	5	0	6	-1	0	-5	8
5	{2, 6}	2	0	-7	-1	0	-5	-3
6	{4, 6}	4	0	-7	-1	-13	-5	-3
7	{5, 6}	5	0	-7	-1	-13	-18	-5
8	{2, 6}	2	0	-20	-1	-13	-18	-16
9	{3, 4, 6}	4	0	-20	-12	-26	-18	-16
10	{3, 5, 6}	5	0	-20	-12	-26	-31	-18

Table 3: Iterations of the shortest path algorithm for Network 3

The iterations of the shortest path algorithm are reported in Table 3. Note that the lower bound is

$$(m - 1) \min_{(i,j)} c_{ij} = 5 \cdot -6 = -30.$$

We observe that the algorithm does not converge. There is a negative cost cycle $2 \rightarrow 4 \rightarrow 5 \rightarrow 2$. Its cost is -13 . The labels of these nodes decrease over and over. The algorithm stops when the lower bound has been reached. Note that applying Dijkstra's algorithm on this network would generate an infinite number of iterations. Indeed, Dijkstra's algorithm does not check for negative cost cycle, and stops when the set $\mathcal{S}$ is empty, which is never the case here.